

A Conditional and Empirical Analysis of Budget Deficits and Economic Growth in Canada

Christian P. and group

University of Toronto Mississauga

ECO202Y5: Macroeconomic Theory and Policy

Dr. Michael Ho and Dr. Masoud Anjomshoa

April 3, 2026

Introduction

Rationale

Fiscal policy is deeply important to GDP growth in every country, including in Canada. While modern-day Canada has existed for over 150 years, our analysis will start from the year 1980, as about 40 years of data is sufficient to analyze Canadian fiscal policy while not going too far into the past. The federal government of Canada runs a budget deficit more often than it runs a surplus, which is a comparable situation to most other developed countries (OECD, 2023). Between the fiscal years of 1980-1981 and 2024-2025, the federal government ran a deficit for 34 years and a surplus for 11 years (Department of Finance Canada, 2025). Governments run deficits for a variety of reasons, primarily to stimulate GDP growth. While in most years it is not practical for the federal government of Canada to run a fiscal surplus or even balance their budget, deficit spending does have the potential to cause harmful fiscal effects. In the early 1990s, the federal government of Canada experienced this as interest payments on public debt, which is accumulated from deficits. Interest on debt took up a total of nearly 30% of total federal government spending (about 6% of GDP) in both the 1990-1991 and 1996-1997 fiscal years (Department of Finance Canada, 2025). This is clearly not economically productive spending, as the interest payments go to wealthy individuals and institutions. In response to this serious fiscal challenge, the Chrétien federal government was effectively forced to slash spending across the board (Veldhuis and Clemens, 2014). Total program expenses fell from 17% of GDP in 1992-1993, to 11.8% of GDP in 1999-2000 (Department of Finance Canada, 2025). As a result, interest payments as a percentage of total federal government spending fell to decrease to 16% by 2005-2006 (about 2.4% of GDP).

This report will analyze the extent to which fiscal policy affects GDP growth, as it can be difficult to break down if an increase in output is the result of a change in fiscal policy or some other factor. It is also vital to consider how GDP growth from expansionary fiscal policy changes directly leads to an improvement in quality of life, particularly since consumption is the component of GDP most closely connected to household well-being.

Goal

This paper examines the relationship between government budget deficits and economic growth in Canada. Its main purpose is to determine whether deficit spending contributes to economic growth in the short run and whether it supports sustainable growth in the long run. Rather than treating all deficits as either always overall beneficial or harmful, the paper will evaluate the conditions under which deficit spending may stimulate output, and the conditions under which it may create economic costs. Additionally, this paper will examine which components of government spending are the most effective, by analyzing fiscal multipliers. This is very relevant today as the new Canadian government has drastically moved the focus of fiscal policy, by increasing spending in some areas and cutting spending in others, while cutting certain taxes. It will also consider the federal government's tax policy over time, as increases and decreases in tax rates are another component of fiscal policy. Finally, this paper will attempt to determine if there is causation and not solely correlation between government policies and GDP growth.

Methodology

The report will use combine mixed strategies to specific analyze how countries running budget deficits impact economic growth. For accuracy, only data from official sources such as Statistics Canada, the International Monetary Fund, the Federal Reserve and the World Bank will be used for the analysis of growth. Throughout our analysis, we take teachings from the Pearson's textbook, *Macroeconomics 9th Edition*, to help examine previous major economic shocks in Canada. We will also analyze sector specific components of the Canadian economy since some sectors are more sensitive. We will use theory using macroeconomics models such as the IS-LM model and the Solow growth model and apply them to real world data.

To investigate the relationship between government deficits and GDP growth, this study makes use of a combination of descriptive statistics, correlation analysis and regression techniques on data from 1980 to 2024. The dataset includes real GDP growth, government balance (as a percentage of GDP) and additional macroeconomic controls.

Government fiscal position was measured using the general government balance (as a percentage of GDP). For ease of interpretation, the deficit variable was generated using the following relationship:

$$\text{Deficit} = -(\text{government balance})$$

so that higher values correspond to larger deficits.

To account for potential confounding macroeconomic factors, additional variables were used (specifically inflation and oil prices). These variables were converted into annual averages before being merged with the main dataset by year.

In terms of data processing, descriptive statistics and time-series plots were used to analyse trends in fiscal balance and economic growth over time. Afterwards, scatterplots and Pearson correlation coefficients were used to examine the correlational relationship between deficits and GDP growth.

Beyond this, Ordinary Least Squares (OLS) regressions were conducted. This technique acts as a standard estimator for linear relationships. Two regressions were done: (1) same-year growth regressed on the deficit and (2) next-year growth regressed on the deficit to check if deficits occurred during downturns and its potential effects on future recovery. The equations for the regression lines are:

- (1) Same year growth: $g_t = \alpha + \beta \cdot \text{deficit}_t + \gamma \cdot g_{t-1} + \delta'Z_t + \varepsilon_t$
- (2) Next year growth: $g_{t+1} = \alpha + \beta \cdot \text{deficit}_{t+1} + \gamma \cdot g_t + \delta'Z_{t+1} + \varepsilon_{t+1}$

Where:

g_t : GDP growth at time t

g_{t+1} : GDP growth at time t+1

Z_t : vector of other macroeconomic factors affecting growth

$\varepsilon_t, \varepsilon_{t+1}$: error terms

deficit_t : government deficit at time t

α : intercept

β : effect of the deficit on growth

γ : effect of past growth

δ' : effect of control variables

Due to the time series manner of the data, there is a possibility for the model's errors to be autocorrelated. Furthermore, the variance of error terms over time is typically not constant. Due to this, OLS does not act as the most efficient estimator which increases the risk of making an incorrect conclusion.

Since OLS assumes errors are independent and calculated standard errors may be biased, the Newey-West standard errors (a heteroscedasticity and autocorrelation consistent estimator) is used in the time-series regression to produce reliable statistical inference. The Newey-West adjusts OLS standard errors to account for heteroscedasticity and autocorrelated residuals.

The analysis also includes a comparison between deficit and non-deficit years, including a "large deficit" indicator (where large is defined as a balance $< -3\%$ of GDP). To assess structural changes, the sample is split into pre-1991 and post-1991 periods and regressions are re-estimated separately for each subsample.

Moreover, Solow's perspective will be used to examine the long-run effects of Canada's budget deficits and how they may impact economic growth. Developed by Robert Solow, the model attempts to determine economic output through capital, labor, and technological progress (Solow, 1957). From a Solow framework, fiscal deficits support long-run economic growth only if they finance productive investment that increases capital accumulation and productivity (Blanchard, 2024).

To support the Solow framework, this analysis uses historical Canadian data from 1980 to 2024, drawn from the Department of Finance Canada's 2025 Fiscal Reference Tables (Department of Finance Canada, 2025) and 1996-2023 statistics from the World Bank (World Bank, n.d.), to investigate the relationship between deficits and productive investments like research and development (R&D). Two types of data from the 2025 Fiscal Reference Tables and a third dataset from the World Bank on Canadian R&D expenditure (% of GDP) will be referenced. The first type of data is the budgetary surplus or deficit as a percentage of GDP, which shows the federal government's fiscal spending relative to the size of GDP. The second type of data is the gross public debt charges as a percentage of revenues. According to Public Services and Procurement Canada (2025), public debt charges "include effective interest calculated on market debt... as well as interest on obligations for public sector pensions and other employee and veteran future benefits." This measure shows the proportion of government revenue allocated to servicing public debt, which is useful because it shows the fiscal expenses created by debt over time and helps assess whether rising debt charges may reduce the government's ability to finance productive investments in the long run.

Analysis

The descriptive statistics provide crucial context for interpreting the results. As shown in Figure 2, Canada's average GDP growth rate over 1980 to 2024 is around 2.31%. At the same time, the average fiscal deficit calculated in the sample makes up approximately 2.88% of GDP. The standard deviation in the balance (3.6953) suggests the presence of large macroeconomic shocks, including recessions and crisis periods. Theoretically, these patterns are consistent with Keynesian economics where governments adopt countercyclical fiscal policies to stabilize fluctuations in the business cycle.

Correlation analysis demonstrates a consistent trend: deficits are slightly negatively correlated with current growth, while positively correlated with growth in the following year (Figure 4 and Figure 5). This pattern is consistent with the mechanisms of automatic stabilizers. Automatic stabilizers are features of government budget which mediate economic cycles without requiring new legislation. Fiscal deficits tend to increase during economic downturns due to a decrease in tax revenue and a rise in government spending. As economic conditions recover over time, growth improves in later periods.

Furthermore, Figure 7 shows that years with large deficits (balance $< -3\%$) have lower average growth ($\sim 1.89\%$) than other years ($\sim 2.68\%$), but this difference is not statistically significant ($p \approx 0.28$), which means there is only weak evidence of systematic differences.

The sign change over time is a result of the main macroeconomic process of countercyclical fiscal policy. During recessions, economic activity slows down, leading to automatic stabilizers like falling tax revenues and rising transfers. These make deficits bigger, and discretionary fiscal responses may make them even bigger. During recoveries, growth picks up after these times of fiscal expansion. This pattern closely follows Keynesian theory, which says that fiscal policy is important for stabilizing overall demand, especially when the economy is slow (Keynes, 1936; Auerbach & Gorodnichenko, 2012). Neoclassical frameworks, on the other hand, show that long-term deficits can slow down medium-term growth by crowding out, lowering savings, and raising risk premia.

It is important to consider whether and to what extent the government can realistically source funding and how expectations about interest rates coexist with growth (OECD, 2012; Blanchard, 2019). As such, the findings from the regressions should be viewed as reduced-form relationships between the variables. This limitation arises from some identification challenges. First, reverse causality is an issue because of the nature of economic slumps, which tend to make fiscal deficits worse by reducing tax receipts and increasing expenditures. There is also the matter of denominator problems in that contractions in GDP lead to higher ratios. Finally, fiscal policy is an endogenous process often employed with expansionary measures often implemented in response to weak economic conditions (IMF Fiscal Monitor, 2015).

The analysis reveals a strong empirical pattern aligned with countercyclical fiscal behavior, rather than indicating that deficits intrinsically induce changes in growth. The negative

contemporaneous relationship illustrates recession dynamics, whereas the positive lagged relationship signifies subsequent recovery. This is evident through Figure 6 which shows that average GDP growth in deficit years is approximately 1.95% compared to 3.12% in non-deficit or surplus years. This aligns with business cycle theory, where governments would increase spending or reduce taxes during recessions to support aggregate demand.

Structural break analysis (pre- vs. post-1991) backs up this interpretation even more. The variations in how deficit affects economic growth implies that Canada's fiscal policy has evolved over the years. In the pre-1991 period, the deficit coefficient is not significant; however, it turns out to be negative and significant in the post-1991 period. As a result, we conclude there is substantial evidence to suggest that deficits play a critical role in stabilizing the economy over the short term, and that automatic stabilizers and discretionary fiscal policies work hand in hand to restore buffers and ensure sustainability. Furthermore, the differences pre- vs post-1991 reflects changes in Canada's fiscal framework where the regime became characterized by formal inflation targeting, deficit elimination and expenditure control (Champagne and Sekkel, 2018).

In Figure 8 and Figure 9, regression analysis for same-year growth provides more evidence of the countercyclical nature of fiscal policy. The introduction of control variables like inflation, the influence of COVID, and structural reforms after 1995 resulted in a negative and statistically significant coefficient for the deficit variable. The data shows that higher deficits result in lower growth rates in the same year, implying that deficits rise when the economy is experiencing a downturn. This observation should not be interpreted in a way that the deficit leads to lower growth, but rather that the deficit reacts to an economic downturn. However, in the case for next-year regression, there is a positive correlation between deficits and growth rates (Figure 10, Figure 11 and Figure 12). This suggests that periods with high deficits are followed by improved economic performance. This result may indicate that there might be fiscal multipliers in the economy. However, it can be attributed to the stage of the business cycle when deficits reach their highest levels during the recession and growth takes place afterward.

The introduction of inflation as a control variable helps add another dimension to the understanding of macroeconomic behavior. Although inflation has no major effect on current-year growth, there is evidence of a negative correlation of statistical significance in terms of its influence on next-year growth. In any case, these results serve as an appropriate framework when evaluating contemporary fiscal policy.

The federal government's budget deficit is currently at \$78 billion for the 2025 to 2026 fiscal year, projected to gradually decrease to \$57 billion by the 2029-2030 fiscal year (Budget 2025). Government set a fiscal anchor of maintaining a declining deficit-to-GDP and debt-to-GDP ratio to ensure disciplined fiscal management for future generations. However, it is not guaranteed these goals will be achieved depends on the rate of future GDP growth (Budget 2025).

Opposition parties had different views of what the government's fiscal policy should be. The government rejected Conservative Party opposition demands to run a smaller deficit of \$42 billion, as they argued that that would result it cuts to "vital social programs and all the capital investments needed for Canada's future" (Hudes, 2025). The 2025 Conservative Party platform promised to cut spending by about the same amount of the Liberal Party but instead proposed to

use the savings mostly cut taxes rather than increase capital spending (CBC/Radio Canada, 2025). The New Democratic Party opposed the government's spending cuts intended to decrease the size of the budget deficit (Zhu, 2025), as their platform promised increasing certain taxes instead (CBC/Radio Canada, 2025). Bloc Quebecois party opposed the budget because they believed that it did not increase health transfers to the provinces and Old Age Security sufficiently (Steven, 2025). These political disagreements about fiscal policy are important to consider, as fiscal policy is fundamentally formulated by elected lawmakers in Canada. This reveals that the optimal fiscal policy for society may inherently be subjective, as the people mostly in charge of it disagree about how much the government should spend in different area and at what rates different taxes should be set at.

The government has the goal of not only using public spending to increase economic growth but also plans to use this spending to encourage and stimulate private investment (Budget 2025). While this could function as intended, the government still needs to consider the crowding out effect, as the increased deficit could decrease investment from the private sector (Blanchard, 2024).

While this increase in capital spending should stimulate economic growth, it will also be mostly funded by deficit spending which needs to be paid back with interest. Interest on public debt was over \$53 billion in 2024-2025, which is double of what it was in the 2021 to 2022 fiscal year at about \$24.5 billion and only slightly higher than in 1995-1996 at \$49.4 billion. However, since Canada's GDP is significantly higher now than in the 1995-1996 fiscal year, it is far more fiscally manageable, as it represents only 1.7% of GDP compared to about 5.9% of GDP in the 1995-1996 fiscal year (Department of Finance Canada, 2025).

Notably, the government focused much of the new spending on a massive increase in defence spending, with about \$80 billion in new funding over the next 5 years (Budget 2025). This could be seen as hedging risk against an attack, and therefore the resulting damage to economic capital that one would incur. However, the government is attempting to use this increase to stimulating economic growth, as they increased troop pay and formulated a Defense Industrial Strategy to attempt to manufacture the capital that they purchase primarily from Canadian firms (Budget 2025).

Canada has had significantly lower government bond yields (cheaper for the government to borrow to fund deficit spending) than the US and UK governments since 2023. When Budget 2025 was published, Canada's 10-year government bond yield was 3.1%, while it was 4% in the US and 4.4% in the UK (Budget 2025). However, the rate increased since the 2025 budget slightly increased to 3.22 (Bank of Canada, 2026). This shows that the Canadian government can still borrow for cheaper than for other wealthy countries. Since the target policy interest rate has slightly decreased from 2.75% to 2.25% since much of the federal government's deficit funded expansionary fiscal policy began, that decrease amplified the effect of the fiscal policy and decreased the currency exchange rate, as depicted in figure 17.

The Parliamentary Budget Officer also estimated the fiscal multipliers in June 2025 for six different expansionary fiscal multipliers for five proceeding years. For one year after the fiscal policy change, "housing investment measures" has a multiplier of 0.9, "infrastructure investment measures" has a multiplier of 0.7, "business investment measures" has a multiplier of 0.6, "measures for modest and low-income households" has a multiplier of 0.7, "personal income tax measures" has a multiplier of 0.3, and "corporate income tax measures" has a multiplier of

0.2. For five years after the fiscal policy change, housing and infrastructure investment measures have the same multiplier as one year after, but business investment measures' fiscal multiplier slightly increases to 0.7 while modest and low-income household measures' fiscal multiplier slightly decreases to 0.6. Finally, the fiscal multipliers for personal and corporate income tax measured significantly increases to 0.7 and 0.5, respectively (Parliamentary Budget Officer, 2024). This explains why the Canadian government is heavily focusing on capital investment and less on tax cuts, as it generally has a higher multiplier and has a shorter lag period before the impact on GDP is realized.

Building on this policy rationale, it is important to consider whether deficit-related strategies are beneficial under different states of the economy (considering the levels of demand, output, and inflation). The following paragraphs discuss how different economic conditions respond to a budget deficit. The findings addressed are based on data extracted from: Statistics Canada's table on Canada's unemployment rate, participation rate, and employment rate, as well as information from the Pearson's textbook - Macroeconomics 9th Edition.

Previously, we have observed that there tends to be an increased budget deficit during weak economic conditions. This is because a budget deficit tends to have a positive effect on the economy when the economy is in a recession, when unemployment is high and when inflation is low. During a recession, actual output falls below potential output (i.e. economy runs under its full capacity). As firms begin to produce less in such conditions, it results in an increase in layoffs, fewer job opportunities which results in higher unemployment. This causes a decline in household income, as well as a decrease in consumer spending. Overall, aggregate demand continues to drop. When inflation is low, this is also a sign that demand is weak. When consumer spending is high, firms are able to increase their prices. Accordingly, inflation rises too. However, when demand is low, firms are unable to raise prices and inflation reflects this strain.

Under these circumstances, a budget deficit plays an important role in stabilizing the economy. By increasing government spending, this directly pushes aggregate demand upwards. Consequently, implementing a budget deficit will help stimulate production and reduce unemployment. To meet the higher demand, firms will produce more. Furthermore, part of government spendings is used towards infrastructure and social programs which help create jobs and increase household incomes.

This case was demonstrated during the 2008 financial crisis. Figure 15 illustrates that Canada was running a budget surplus prior to 2008. In the years leading up to the crisis, unemployment remained relatively stable, and GDP growth was slow but still rising. However, the market crash in 2008 resulted in the unemployment rate increasing drastically and GDP growth declining, as shown in Figure 8 and Figure 2. In response to the financial crisis, the government ran budget deficit to help increase demand and stimulate the economy. By doing so, the rise in unemployment slowed and GDP growth was able to return to similar levels from before the crisis. Similar findings can be seen during 2019-2021 due to the COVID-19 pandemic. Before the pandemic began the government was running relatively small budget deficits (Department of Finance Canada, 2025). When lockdown was implemented, however, many people were out of work and consumer spendings declined, leading to a drastic decrease in GDP

growth. To counter this shock, the government ran a large budget deficit, which helped in restoring the economy.

Though large budget deficits can play a crucial role in stabilizing the economy in times of crisis, these effects are not universally beneficial. It is important to consider conditions under which running a deficit may lead to adverse outcomes. This happens when the economy is at or close to operating at full capacity, when inflation is high, and when government debt grows too large. When the economy is already at or close to running at maximum capacity, this means actual output is at or near potential output and most resources, such as labour and capital, are already in use. By running a budget deficit, this creates additional demand that the economy cannot support. In this scenario, there are not enough workers or resources to meet demand. As a result, instead of producing more, prices increase, in turn, inflation increases. Therefore, rather than promote economic growth, the budget deficit results in higher inflation.

As we can see, budget deficits can be effective tools for stabilization if implemented under the right conditions. To maximize economic growth, it is important to consider the state of the economy at the time. When aggregate demand is low, running a budget deficit can promote economic activity and support employment. Whereas, when aggregate demand is high, it is best to avoid running a budget deficit to prevent the economy from overheating and increasing inflation.

Although budget deficits may contribute to short-run economic stability, short-run stimulation does not always translate into long-run growth. The data from Figure 14 shows that the federal gross public debt charges generally decline over the long run, reaching a low point of 5.9% around the early 2020s. However, after COVID-19, this ratio began to rise, suggesting that, despite debt-servicing costs becoming more manageable for a while, recent deficits and higher borrowing costs have started to increase the fiscal burden again (Department of Finance Canada, 2025). Glancing at Figure 15, which shows Canada's budgetary surplus or deficit as a percentage of GDP, the federal government has run deficits for most of the past four decades. One exception was the period roughly from 1996 to 2007, when Canada achieved a series of mild budget surpluses. This did not last long, as the government returned to deficit spending during the 2008-09 financial crisis. Thereafter, the budget balance fluctuated around roughly 0% to -1.7% of GDP until 2020–21. During the COVID-19 pandemic, the deficit rose sharply to about -14% of GDP, which was by far the largest in the sample (Department of Finance Canada, 2025). Together, these figures suggest that the Canadian government has continued borrowing over time, and the recent rise in debt charges shows that this fiscal burden may become more difficult to manage in the future.

Additionally, Figure 16 presents World Bank data on Canada's research and development expenditure as a percentage of GDP. The data show that R&D spending rose steadily and reached a peak of about 2.02% in the early 2000s. After this, it followed a mild downward trend, falling to around 1.68% in 2017. Despite spending increase for a short period, including a rise to about 1.93% in 2020, it later declined once again (World Bank, n.d.). Thus, the data suggest that Canada's R&D spending has remained within a relatively narrow range over time, fluctuating between 1.6% and 2.0% of GDP. From a Solow perspective, this suggests that Canada has continued support for innovation, but not at a point that clearly shows a strong long-run rise in technological progress.

Canada's data suggest that persistent borrowing has created a growing fiscal burden, while this section does not show equally strong gains in capital accumulation or productivity. This matters because long-run growth depends not only on capital accumulation but also on technological progress. R&D supports innovation and productivity, which is one of the main ways in which deficit spending could improve future output. But, in Canada's case, these long-run gains appear limited. Instead, the evidence suggests that continued deficits may reduce the government's ability to invest in productive sectors over time if more revenue is absorbed by debt charges.

Conclusion

Using descriptive statistics, correlation analysis and regression techniques to investigate the relationship between government deficits and GDP growth, this study has shown that deficits at time t are weakly negatively correlated with growth at time t but are positively correlated with growth at time $t+1$. Furthermore, we found that an observable correlation related to countercyclical fiscal behavior. Through structural break analysis, changes in Canada's fiscal system means that the relationship between deficits and growth is not stable across periods. The findings illustrate business cycle dynamics instead of causal fiscal multipliers, showcasing a critical and methodologically robust interpretation of macroeconomic data.

In the process of creating the regression model, we originally intended to include control variables such as oil and inflation. However, data for oil prices were limited to post-2021. This means that we were unable to make use of oil prices as a control for the full sample without reducing the number of observations. Due to this, oil prices were excluded from the main specifications.

To advance beyond reduced-form analysis, subsequent research could integrate more robust identification strategies, including the disaggregation of fiscal components (investment versus transfers), or advanced data processing methods such as vector autoregressions (VARs) or localized projections. The primary contribution of this analysis is not only the identification of empirical relationships but also the acknowledgment of their limitations. The findings illustrate business cycle dynamics instead of causal fiscal multipliers, showcasing a critical and methodologically robust interpretation of macroeconomic data.

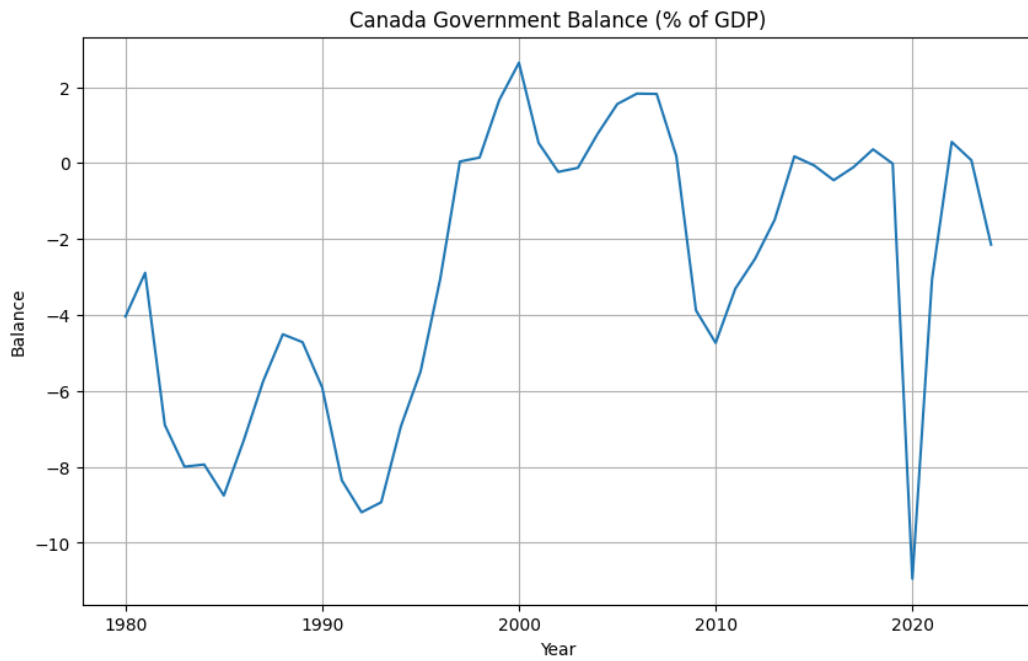
When looking at empirical data from the 2008 financial crisis and the COVID-19 pandemic, we can see that when actual output drops lower than potential output, this causes high unemployment and low inflation. Our analysis demonstrates that in these cases a budget deficit supports the restoration of the economy. However, our research fails to consider alternative factors that contribute to the economy's recovery. For example, a major cause for the decline in GDP growth during COVID-19 pandemic was the implementation of lockdown restrictions, limiting people's ability to spend. Though, once these restraints were lifted and people were able to settle back into their old routines, this allowed for consumption to increase back to similar levels from before the pandemic. Therefore, it is important to note that although the government's decisions play a role in maintaining stable economic growth, other factors, such as global health conditions and consumers incentives, influence the outcomes of the economy.

One goal that was not possible to fully achieve was proving causation rather than just correlation. Since it is clearly not possible to conduct an isolated experiment on then the entire Canadian economy. Another drawback was data inconsistencies. Canada's World Bank data for R&D was only available starting in 1996. This indicates that the long-run comparison is not entirely consistent across all variables and is based on various time spans. Nonetheless, the information is still helpful in demonstrating that Canada's investment in R&D stayed within a relatively small range throughout time.

Overall, it is vital for policymakers to be cautious about deficit spending due to the risks of uneconomically productive high interest charges while still generally pursuing it when necessary to stimulate economic growth, and to attempt to ensure that it maximizes the overall wellbeing of society.

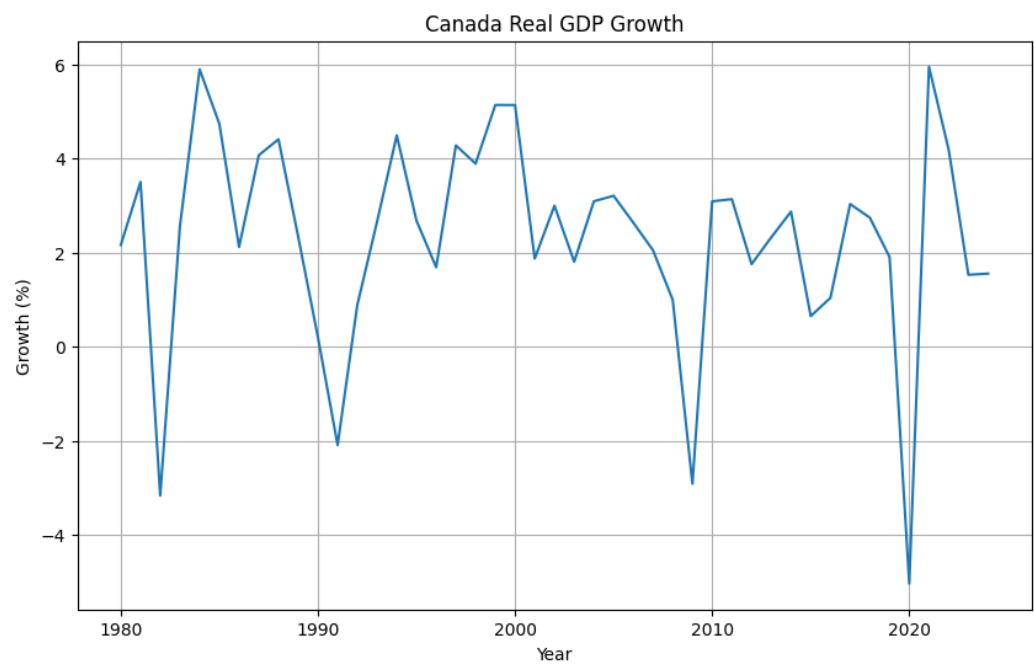
Appendix

Figure 1



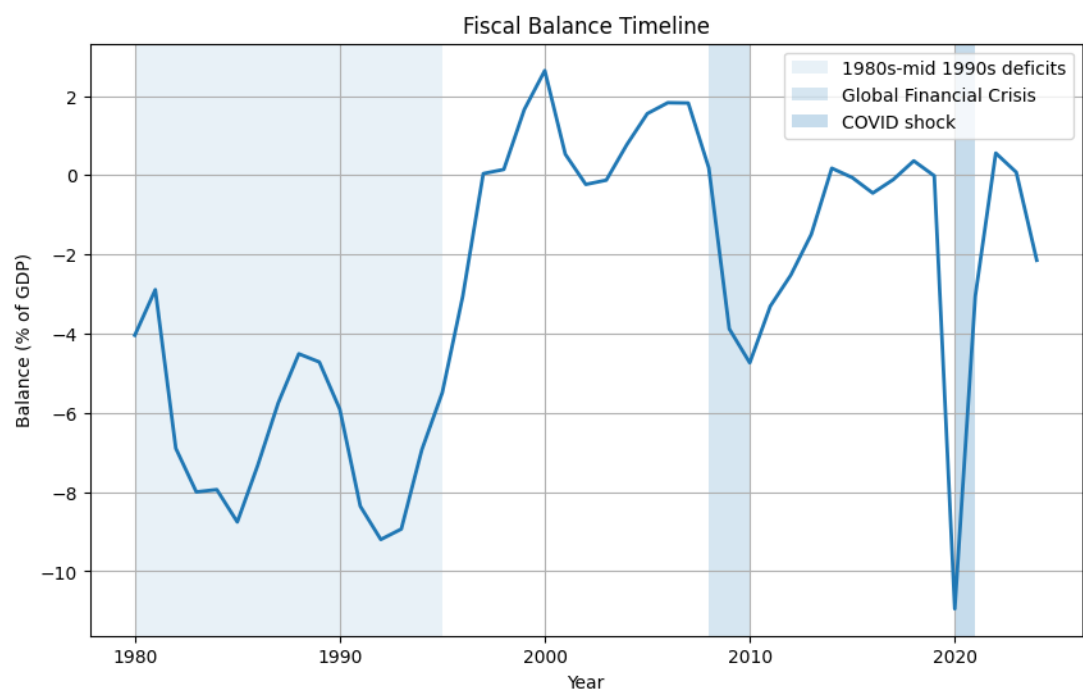
Source: Federal Reserve Bank of St. Louis (2025)

Figure 2



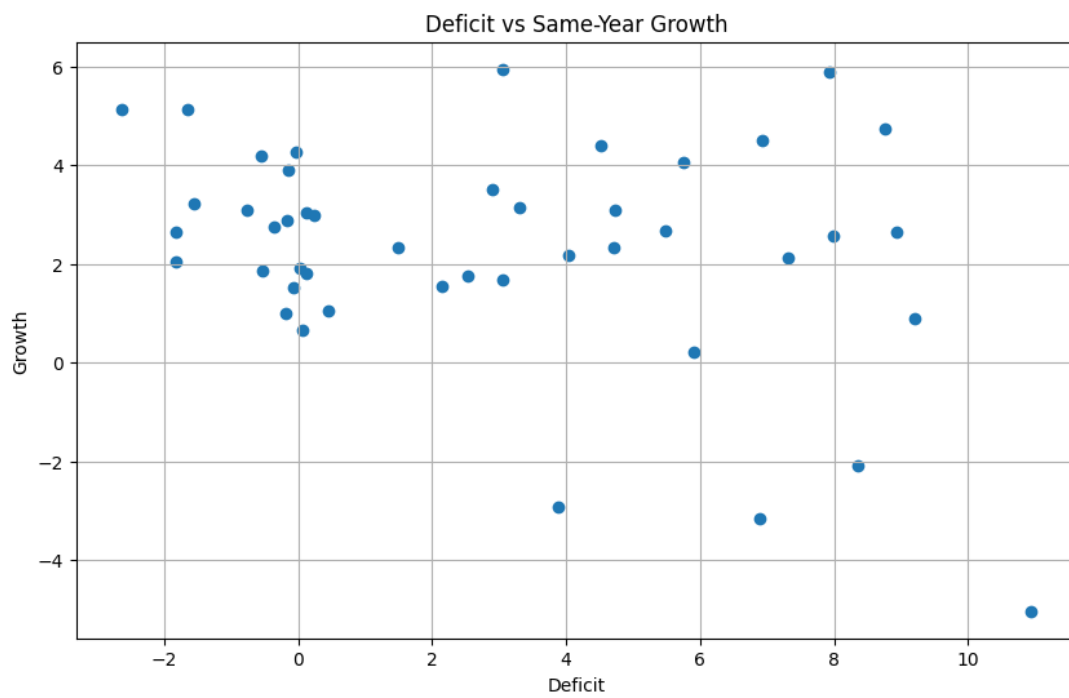
Source: Federal Reserve Bank of St Louis (2025)

Figure 3



Source: Federal Reserve Bank of St. Louis (2025)

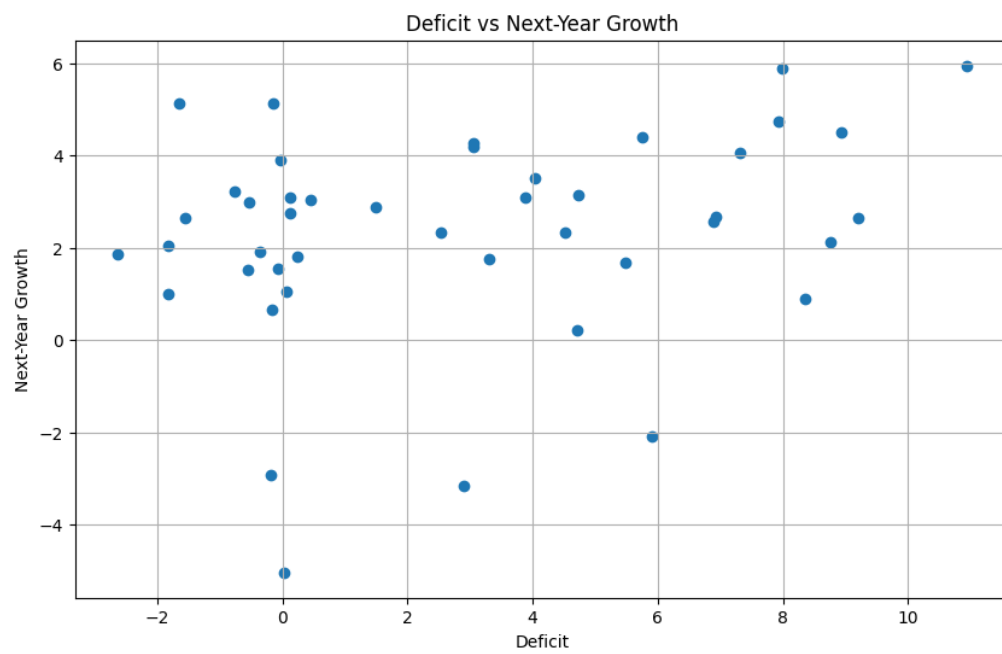
Figure 4



Same-year correlation: -0.304

Source: Federal Reserve Bank of St. Louis (2025)

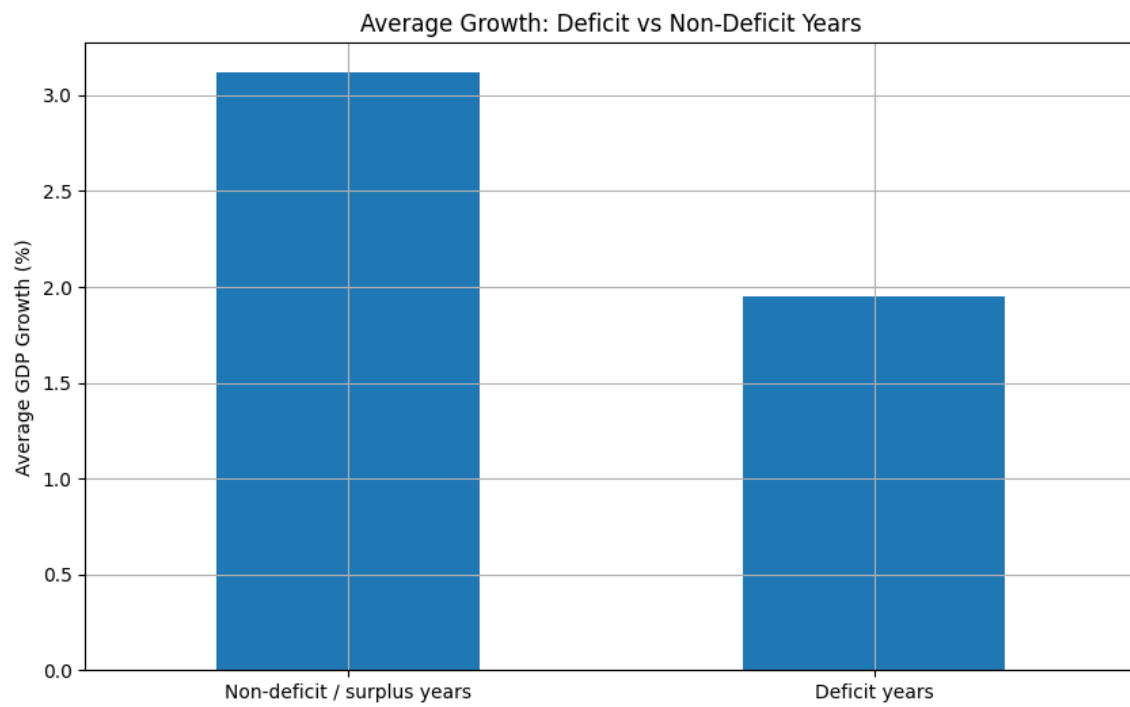
Figure 5



Next-year correlation: 0.253

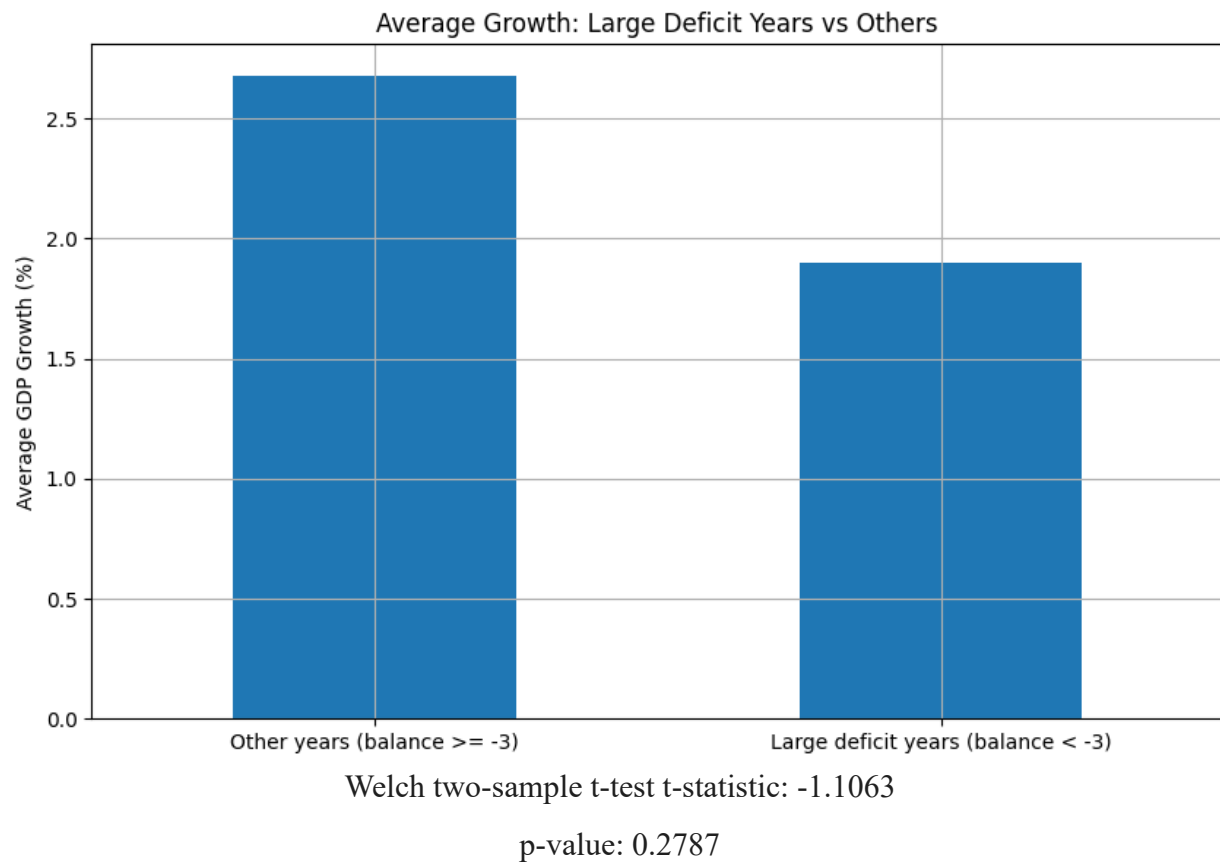
Source: Federal Reserve Bank of St. Louis (2025)

Figure 6



Source: Federal Reserve Bank of St. Louis (2025) & Federal Reserve Bank of St. Louis (2026)

Figure 7



Source: Federal Reserve Bank of St. Louis (2025) & Federal Reserve Bank of St. Louis (2026)

Figure 8

```
=====
MODEL: Same-year growth ~ deficit
=====

Dependent variable: growth
Independent variables: ['deficit']
Usable observations: 45

                                OLS Regression Results
=====
Dep. Variable:                  growth    R-squared:                        0.093
Model:                        OLS        Adj. R-squared:                   0.072
Method:                       Least Squares    F-statistic:                     2.268
Date:                        Fri, 03 Apr 2026    Prob (F-statistic):              0.139
Time:                        03:58:54         Log-Likelihood:                  -97.589
No. Observations:              45            AIC:                             199.2
Df Residuals:                  43            BIC:                             202.8
Df Model:                      1
Covariance Type:               HAC
=====

               coef      std err          z      P>|z|      [0.025      0.975]
-----
const          2.8461      0.321       8.861     0.000       2.217      3.476
deficit        -0.1851      0.123      -1.506     0.132      -0.426      0.056
=====

Omnibus:                4.923    Durbin-Watson:                1.658
Prob(Omnibus):           0.085    Jarque-Bera (JB):              3.701
Skew:                   -0.570    Prob(JB):                      0.157
Kurtosis:                3.821    Cond. No.                      6.03
=====
```

Source: Federal Reserve Bank of St. Louis (2025) & Federal Reserve Bank of St. Louis (2026)

Figure 9

```

=====
MODEL: Same-year growth ~ deficit + inflation
=====
Dependent variable: growth
Independent variables: ['deficit', 'inflation']
Usable observations: 45

                        OLS Regression Results
=====
Dep. Variable:          growth    R-squared:                0.094
Model:                  OLS      Adj. R-squared:             0.051
Method:                 Least Squares    F-statistic:            1.140
Date:                  Fri, 03 Apr 2026    Prob (F-statistic):      0.329
Time:                  03:58:54    Log-Likelihood:         -97.552
No. Observations:      45    AIC:                    201.1
Df Residuals:          42    BIC:                    206.5
Df Model:              2
Covariance Type:       HAC
=====

```

	coef	std err	z	P> z	[0.025	0.975]
const	2.7533	0.512	5.376	0.000	1.749	3.757
deficit	-0.1906	0.128	-1.484	0.138	-0.442	0.061
inflation	0.0343	0.152	0.225	0.822	-0.264	0.332

```

=====
Omnibus:                4.780    Durbin-Watson:           1.650
Prob(Omnibus):          0.092    Jarque-Bera (JB):        3.558
Skew:                  -0.561    Prob(JB):                0.169
Kurtosis:              3.801    Cond. No.:               9.44
=====

```

Source: Federal Reserve Bank of St. Louis (2025) & Federal Reserve Bank of St. Louis (2026) & World Bank (2026)

Figure 10

```
=====
MODEL: Next-year growth ~ deficit
=====
Dependent variable: growth_next
Independent variables: ['deficit']
Usable observations: 44

                        OLS Regression Results
=====
Dep. Variable:          growth_next    R-squared:                0.064
Model:                  OLS            Adj. R-squared:         0.042
Method:                 Least Squares   F-statistic:             2.520
Date:                  Fri, 03 Apr 2026 Prob (F-statistic):       0.120
Time:                  03:58:54         Log-Likelihood:          -96.599
No. Observations:      44              AIC:                    197.2
Df Residuals:          42              BIC:                    200.8
Df Model:               1
Covariance Type:       HAC

=====
                        coef    std err          z      P>|z|      [0.025    0.975]
-----
const                1.8721     0.483     3.877     0.000     0.926     2.818
deficit              0.1537     0.097     1.587     0.112    -0.036     0.344
=====
Omnibus:              16.074    Durbin-Watson:           1.663
Prob(Omnibus):        0.000    Jarque-Bera (JB):        18.736
Skew:                 -1.303    Prob(JB):                8.54e-05
Kurtosis:             4.852    Cond. No.:               6.07
=====
```

Source: Federal Reserve Bank of St. Louis (2025) & Federal Reserve Bank of St. Louis (2026)

Figure 11

```
=====
MODEL: Next-year growth ~ deficit + inflation
=====
Dependent variable: growth_next
Independent variables: ['deficit', 'inflation']
Usable observations: 44

                        OLS Regression Results
=====
Dep. Variable:          growth_next    R-squared:                0.148
Model:                  OLS            Adj. R-squared:         0.106
Method:                 Least Squares   F-statistic:             7.729
Date:                   Fri, 03 Apr 2026 Prob (F-statistic):       0.00142
Time:                   03:58:54        Log-Likelihood:          -94.527
No. Observations:       44              AIC:                    195.1
Df Residuals:           41              BIC:                    200.4
Df Model:                2
Covariance Type:        HAC
=====
               coef      std err          z      P>|z|      [0.025      0.975]
-----
const          2.5685      0.562      4.570      0.000      1.467      3.670
deficit         0.1947      0.088      2.204      0.028      0.022      0.368
inflation      -0.2554      0.097     -2.639      0.008     -0.445     -0.066
=====
Omnibus:            15.374    Durbin-Watson:           1.589
Prob(Omnibus):      0.000    Jarque-Bera (JB):        18.325
Skew:               -1.199    Prob(JB):                0.000105
Kurtosis:           5.060    Cond. No.                9.49
=====
```

Source: Federal Reserve Bank of St. Louis (2025) & Federal Reserve Bank of St. Louis (2026) & World Bank (2026)

Figure 12

```

=====
MODEL: Next-year growth ~ deficit + inflation + covid + post_1995
=====
Dependent variable: growth_next
Independent variables: ['deficit', 'inflation', 'covid', 'post_1995']
Usable observations: 44

                                OLS Regression Results
=====
Dep. Variable:          growth_next    R-squared:                0.177
Model:                  OLS            Adj. R-squared:          0.092
Method:                 Least Squares   F-statistic:              23.06
Date:                  Fri, 03 Apr 2026 Prob (F-statistic):       7.79e-10
Time:                  03:58:54         Log-Likelihood:          -93.775
No. Observations:      44              AIC:                    197.6
Df Residuals:          39              BIC:                    206.5
Df Model:              4
Covariance Type:       HAC
=====

```

	coef	std err	z	P> z	[0.025	0.975]
const	2.3288	1.228	1.897	0.058	-0.078	4.735
deficit	0.1785	0.110	1.620	0.105	-0.038	0.395
inflation	-0.2216	0.107	-2.073	0.038	-0.431	-0.012
covid	1.7996	0.603	2.983	0.003	0.617	2.982
post_1995	0.1468	0.931	0.158	0.875	-1.679	1.972

```

=====
Omnibus:                16.111    Durbin-Watson:           1.526
Prob(Omnibus):          0.000    Jarque-Bera (JB):        20.426
Skew:                  -1.201    Prob(JB):                3.67e-05
Kurtosis:              5.317     Cond. No.                45.7
=====

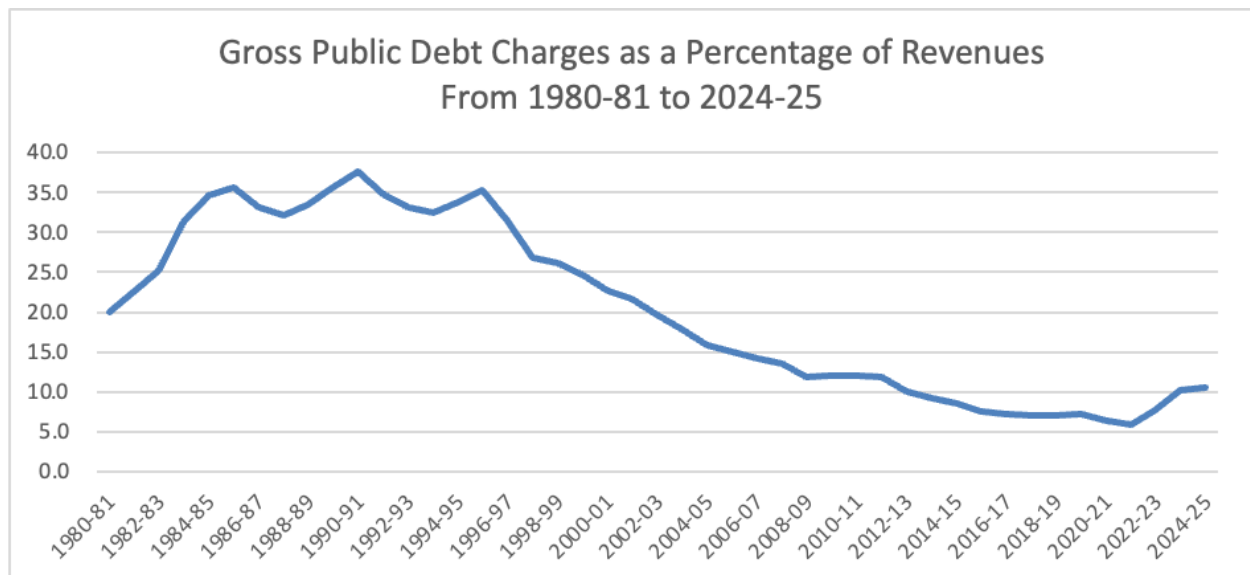
```

Source: Federal Reserve Bank of St. Louis (2025) & Federal Reserve Bank of St. Louis (2026) & World Bank (2026)

Figure 13

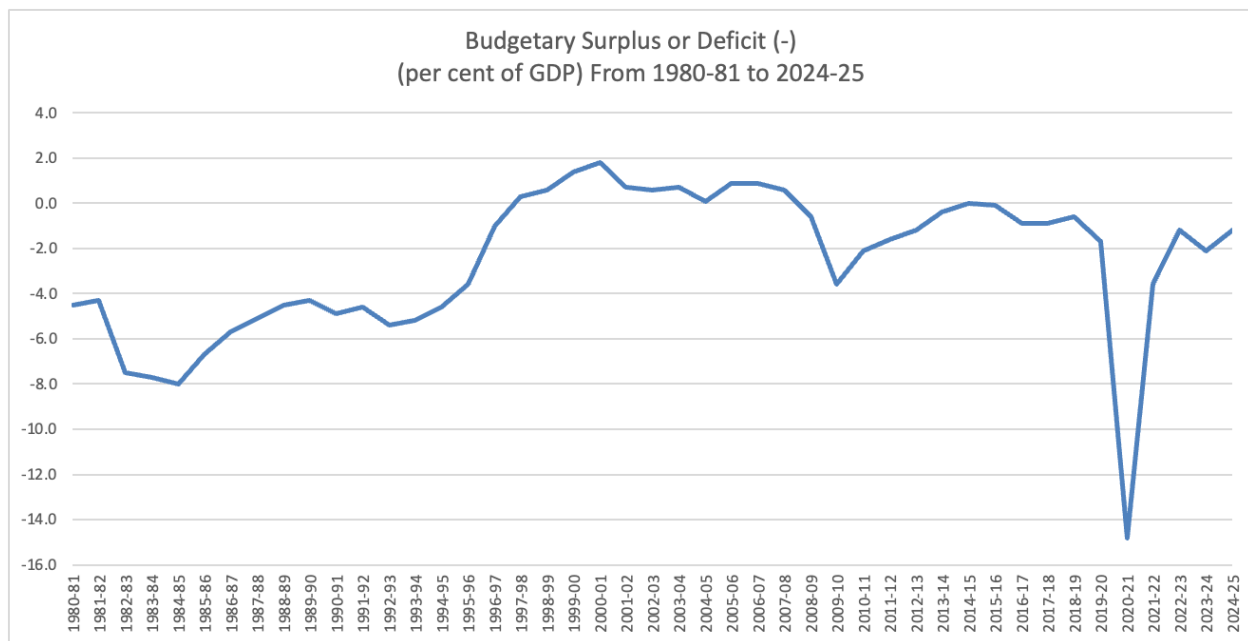


Figure 14



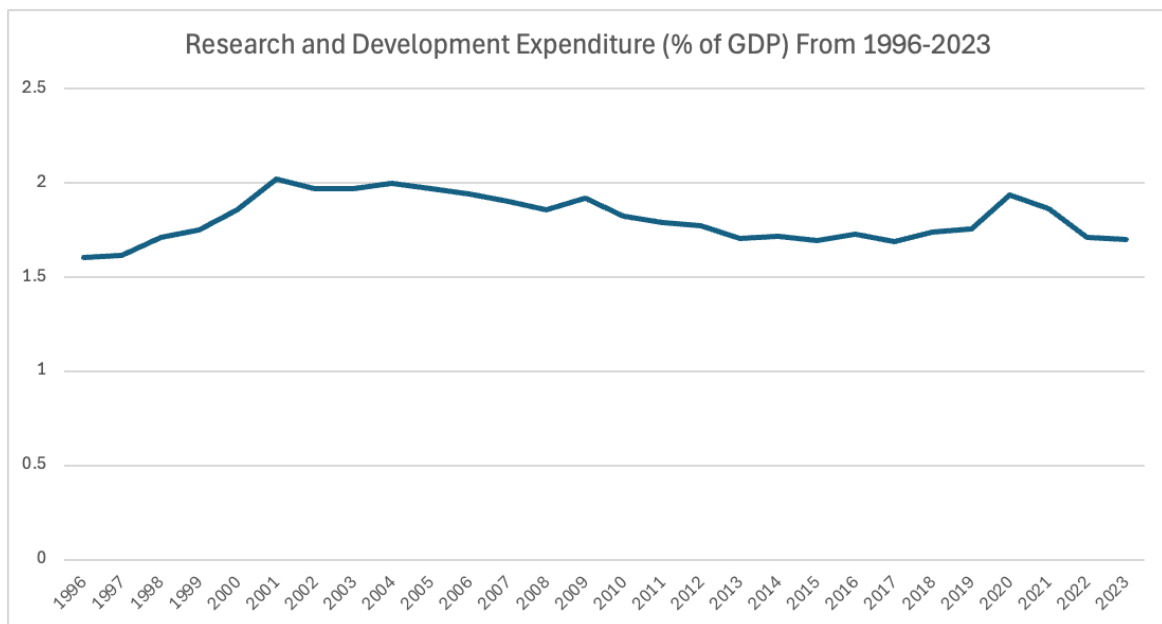
Source: Department of Finance Canada (2025), *Fiscal Reference Tables* (November 2025).

Figure 15



Source: Department of Finance Canada (2025), *Fiscal Reference Tables (November 2025)*.

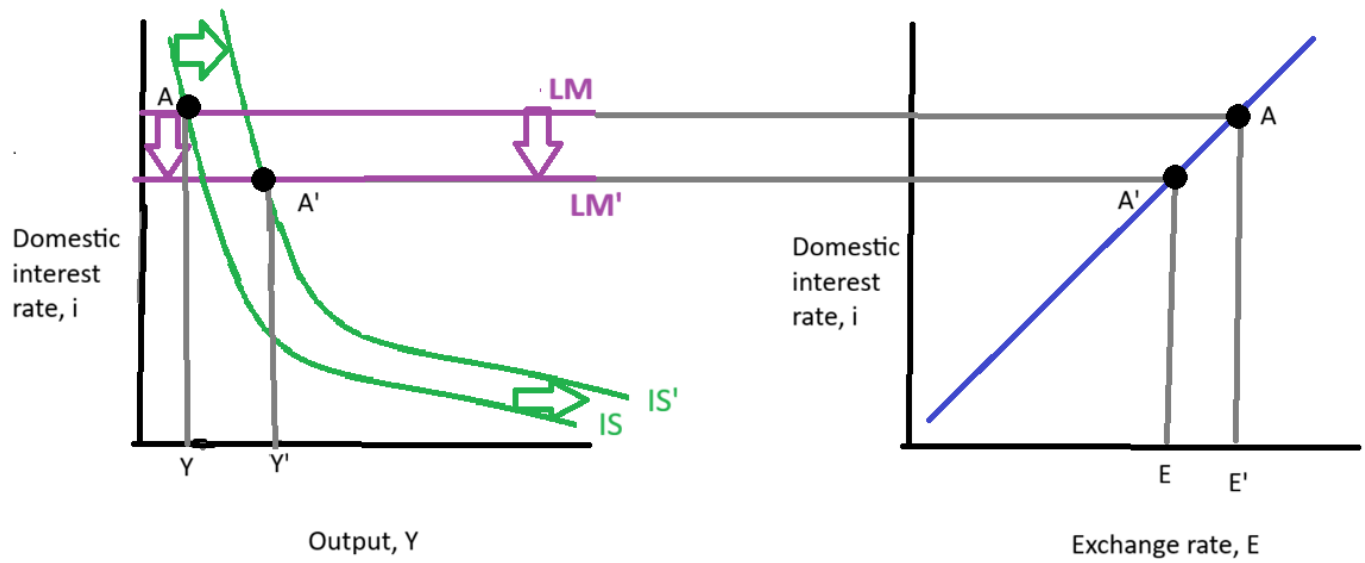
Figure 16



Source: World Bank, *Research and development expenditure (% of GDP) - Canada*

Figure 17

An increase in government spending and a decrease in the interest rate:



Bibliography

Auerbach, A. J., & Gorodnichenko, Y. (2012). *Measuring the Output Responses to Fiscal Policy*. *American Economic Journal: Economic Policy*, 4(2), 1–27. <https://doi.org/10.1257/pol.4.2.1>

Blanchard, O. (2024). *Macroeconomics* (9th ed.) Pearson.

Canadian bond yields: 10-year lookup. Bank of Canada. (2026).
<https://www.bankofcanada.ca/rates/interest-rates/lookup-bond-yields/>

CBC/Radio Canada. (2025). *Compare the election promises of Canada's major parties*. CBCnews. <https://newsinteractives.cbc.ca/features/2025/federal-party-platforms/#federal-2025-taxes>

Champagne, J., & Sekkel, R. (2018). Changes in monetary regimes and the identification of monetary policy shocks: Narrative evidence from Canada. *Journal of Monetary Economics*, 99, 72–87. <https://doi.org/10.1016/j.jmoneco.2018.06.002>

Department of Finance Canada. (2025). *Fiscal Reference Tables (November 2025)*. Government of Canada. <https://www.canada.ca/en/departement-finance/services/publications/fiscal-reference-tables/2025.html>

Federal Reserve Bank of St. Louis. (2025). *General government net lending/borrowing for Canada*. FRED. <https://fred.stlouisfed.org/data/GGNLBACAA188N>

Federal Reserve Bank of St. Louis. (2026). *Gross domestic product for Canada*. FRED. <https://fred.stlouisfed.org/data/CANNGDPRPCH>

Government of Canada. (2016). *History of the Canadian Labour Force Survey, 1945 to 2016*. Statcan.gc.ca. <https://www150.statcan.gc.ca/n1/pub/75-005-m/75-005-m2016001-eng.htm>

Hudes, S. (2025, November 4). *Federal budget forecasts \$78B deficit as liberals shift spending to capital projects*. Investment Executive. <https://www.investmentexecutive.com/news/federal-budget-forecasts-78b-deficit-as-liberals-shift-spending-to-capital-projects/>

International Monetary Fund. (2015, April 15). *Fiscal Monitor, April 2015: Now Is the Time: Fiscal Policies for Sustainable Growth*. IMF.
<https://www.imf.org/en/publications/fm/issues/2016/12/31/now-is-the-time-fiscal-policies-for-sustainable-growth>

Keynes, J. M. (1936). *The General Theory of Employment, Interest, and Money*.
https://www.files.ethz.ch/isn/125515/1366_keynestheoryofemployment.pdf

Legree, D. (2025, November 4). *Conservatives, bloc set to oppose liberal budget, while NDP position remains uncertain - ipolitics*. ipolitics.
<https://www.ipolitics.ca/2025/11/04/conservatives-bloc-set-to-oppose-liberal-budget-while-ndp-position-remains-uncertain/>

OECD. (2012). *Restoring Public Finances, 2012 Update*. OECD Publishing.

Organisation for Economic Co-operation and Development). (2023). *General government deficit*. OECD. <https://www.oecd.org/en/data/indicators/general-government-deficit.html>

Parliamentary Budget Officer. (2024, November 20). *Fiscal multipliers*. <https://www.pbo-dpb.ca/en/additional-analyses--analyses-complementaires/BLOG-2425-007--fiscal-multipliers--multiplicateurs-budgetaires>

Policy interest rate. Bank of Canada. (2026). <https://www.bankofcanada.ca/core-functions/monetary-policy/key-interest-rate/>

Public Services and Procurement Canada. (2025, February 24). *Section 7: Public debt charges—Volume III. Government of Canada*. <https://www.tpsgc-pwgsc.gc.ca/recgen/cpc-pac/2024/vol3/s7/index-eng.html>

Solow, R. M. (1957). *Technical Change and the Aggregate Production Function*. *The Review of Economics and Statistics*, 39(3), 312–320. <https://doi-org.myaccess.library.utoronto.ca/10.2307/1926047>

Statistics Canada. *Table 14-10-0021-01 Unemployment rate, participation rate, and employment rate by type of student during school months, monthly, unadjusted for seasonality [Data Table]*. <https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=1410002101>

Steven, B. L. (2025, October 26). *“right now, we don’t have the votes” to pass federal budget, says liberal House leader | CBC News*. CBCnews. <https://www.cbc.ca/news/politics/liberal-house-leader-budget-votes-9.6953599>

Veldhuis, N., & Clemens, J. (2014, February 5). *Federal Liberals reject the party’s successful pragmatism of the 1990s | fraser institute*. Fraser Institute.
<https://www.fraserinstitute.org/commentary/federal-liberals-reject-partys-successful-pragmatism-1990s>

World Bank. (2026). *World Bank Open Data*.
<https://data.worldbank.org/indicator/FP.CPI.TOTL.ZG?locations=CA>

World Bank. (n.d.). *Research and development expenditure (% of GDP) – Canada. World Development Indicators*. Retrieved April 3, 2026, from <https://data.worldbank.org/indicator/GB.XPD.RSDV.GD.ZS?locations=CA>

Zhu, C. (2025, November 18). *Why two NDP mps abstained from the federal budget vote, and what comes next | CBC Radio*. CBCnews. <https://www.cbc.ca/radio/asithappens/ndp-abstained-from-budget-9.6983105>